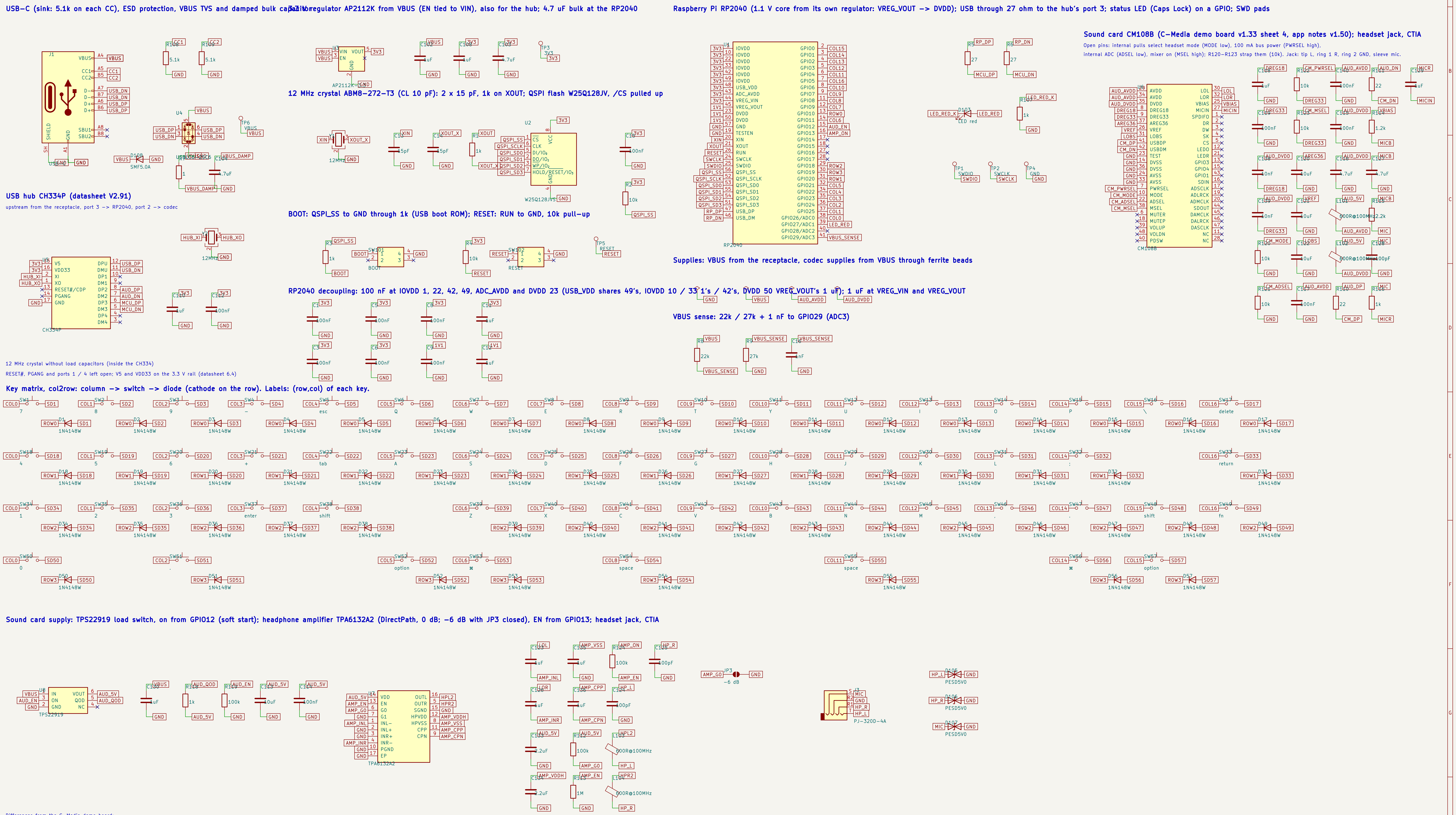
Lira, wired: Raspberry Pi RP2040, USB-C only (no battery), 57 keys, built-in USB sound card

The wireless board (branch main: nRF52840 module, Li-Po) with the radio and the battery taken out; the RP2040 follows Raspberry Pi's 'Hardware design with RP2040' (the Pico's circuit). Differences from the wireless board:

1. USB only, no battery: no Ebyte E73 (nRF52840) module, no Bluetooth, no Li-Po cell, no charger (MCP73831), no power path (B5819W + SI2301), no power switch, no battery divider, no 32.768 kHz crystal, no antenna keep-out.
2. Raspberry Pi RP2040 instead of the module, with the minimum circuit of Raspberry Pi's 'Hardware design with RP2040' (the Pico's): 12 MHz crystal ABM8-272-T3 (the one Raspberry Pi qualified) with 2 x 15 pF and 1k in series on XOUT; W25Q128JV QSPI flash (16 MB) with a 10k pull-up on QSPI_SS; 100 nF (0402) on IOVDD 1, 22, 42, 49, on DVDD 23 and on ADC_AVDD, USB_VDD sharing IOVDD 49's as in the design guide, IOVDD 10 / 33 sharing 1's / 42's through a tie under the package and DVDD 50 VREG_VOUT's 1 uF through the 1.1 V ring under it; 1 uF on VREG_VIN and VREG_VOUT; each within a few mm of its pin on the bottom layer; the internal regulator makes the 1.1 V core supply; TESTEN grounded.
3. USB: hub port 3 goes to the RP2040's USB_DP / USB_DM through 27 ohm series resistors (as on the Pico and in the design guide), instead of the module's D+ / D−.
4. Programming: the RP2040's boot ROM is a USB drive (RPI-RP2) – no bootloader to flash with an SWD probe. A blank board starts in it; BOOT (SW101, QSPI_SS to GND through 1k) held while plugging the cable enters it again; RESET (SW102) pulls RUN low. The SWD pads stay for debugging.
5. 3.3 V regulator AP2112K-3.3 (the same as the wireless board) fed straight from VBUS: 1 uF in, 1 uF + 4.7 uF out; it also supplies the USB hub (V5 and VDD33 on 3.3 V, CH334 datasheet 6.4), so nothing but the regulator, the sound card's switch and the protection sits on VBUS.
6. VBUS protection at the receptacle: SMF5.0A TVS and a 4.7 uF bulk capacitor damped by 1 ohm (hot-plug ringing); VBUS divider 22k / 27k + 1 nF to GPIO29 (ADC3) for the firmware.
7. Status LED (red, Caps Lock) on an RP2040 GPIO, through 1k, as on the wireless board.
8. Test points: SWD (SWCLK, SWDIO, RESET, GND) in one row in front of the RP2040, 3V3 behind it, VBUS; no VBAT / VIN.
9. Sound card: the supply switch U8 is on a GPIO (GPIO12) with a 100k pull-down, as on the wireless board; the headphone amplifier's EN has its own GPIO (GPIO13) through 100k, held low by 1M (the wireless board delays it with an RC from the switch). The firmware turns the sound card on when the host has configured the keyboard, the amplifier 0.2 s later, the amplifier off in USB suspend and as soon as VBUS starts to fall.



USB-C (sink: 5.1k on each CC), ESD protection, VBUS TVS and damped bulk capacitor

3.3 V regulator AP2112K from VBUS (EN tied to VIN), also for the hub; 4.7 uF bulk at the RP2040

Raspberry Pi RP2040 (1.1 V core from its own regulator: VREG_VOUT –> DVDD); USB through 27 ohm to the hub's port 3; status LED (Caps Lock) on a GPIO; SWD pads

Sound card CM108B (C-Media demo board v1.33 sheet 4, app notes v1.50); headset jack, CTIA

Open pins: internal pulls select headset mode (MODE low), 100 mA bus power (PWRSEL high), internal ADC (ADSEL low), mixer on (MSEL high); R120–R123 strap them (10k). Jack: tip L, ring 1 R, ring 2 GND, sleeve mic.

12 MHz crystal ABM8-272-T3 (CL 10 pF): 2 x 15 pF, 1k on XOUT; QSPI flash W25Q128JV, /CS pulled up

USB hub CH334P (datasheet V2.91)

upstream from the receptacle, port 3 –> RP2040, port 2 –> codec

BOOT: QSPI_SS to GND through 1k (USB boot ROM); RESET: RUN to GND, 10k pull-up

Supplies: VBUS from the receptacle, codec supplies from VBUS through ferrite beads

RP2040 decoupling: 100 nF at IOVDD 1, 22, 42, 49, ADC_AVDD and DVDD 23 (USB_VDD shares 49's, IOVDD 10 / 33 1's / 42's, DVDD 50 VREG_VOUT's 1 uF); 1 uF at VREG_VIN and VREG_VOUT

VBUS sense: 22k / 27k + 1 nF to GPIO29 (ADC3)

12 MHz crystal without load capacitors (inside the CH334)

RESET#, PGANG and ports 1 / 4 left open; V5 and VDD33 on the 3.3 V rail (datasheet 6.4)

Key matrix, col2row: column –> switch –> diode (cathode on the row). Labels: (row,col) of each key.

Sound card supply: TPS22919 load switch, on from GPIO12 (soft start); headphone amplifier TPA6132A2 (DirectPath, 0 dB; −6 dB with JP3 closed), EN from GPIO13; headset jack, CTIA

Differences from the C-Media demo board:

1. Supply: VBUS through a TPS22919 load switch (soft start 1.3 ms, output discharge) to AUD_5V, ON from an RP2040 GPIO (on once the host has configured the keyboard): the sound card's capacitors are charged slowly. A hard switch-on would dip VBUS enough to reset the hub (CH334 datasheet 6.1).
2. Headphones: a TPA6132A2 DirectPath amplifier after LOL / LOR (demo board: 100 uF + 39R straight to the jack): about 0 ohm output impedance, no coupling capacitors, 0 dB gain (−6 dB with JP3 closed, for very sensitive earphones). Inputs single-ended through 1 uF (8 Hz); EN from an RP2040 GPIO through 100k (1M to ground): on 0.2 s after the supply, off in USB suspend (2 mA saved) and, when the cable is pulled, as soon as the firmware sees VBUS fall, before the codec loses its supply (no click).
3. Microphone: bias 1.2k + 2.2k (demo board 2.2k + 8.2k, for PC microphones) for phone headsets (Android: 1.8–2.9 V through 2.2k); 1k in series before MICIN against the step when a plug goes in.
4. The demo board's two 10 nF capacitors DVDD – DREG18 and DVDD – DREG33 ('close to IC', for ESD) are fitted; each AVDD pin (29, 34) has its own 100 nF.
5. The amplifier's SGND (pin 15) is grounded at the amplifier (its ground pad and a via into the top ground plane), not on a track of its own to the jack's ground pin as TI suggests (datasheet 10.3): with both layers poured the difference is a few millivolts.

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Title: Lira (wired)

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